

2023 Semester Two (October 2023) Examination Period

Faculty of Science

EXAM CODES: MAT9004
TITLE OF PAPER: Mathematical foundations for data science and AI
EXAM DURATION: 3 hours 10 mins

Rules

During your eExam, you must not have in your possession any item/material that has not been authorised for your exam. This includes books, notes, paper, electronic device/s, smart watch/device, or writing on any part of your body. Authorised items are listed above. Items/materials on your device, desk, chair, in your clothing or otherwise on your person will be deemed to be in your possession. Mobile phones must be switched off and placed face-down on your desk during your exam attempt.

You must not retain, copy, memorise or note down any exam content for personal use or to share with any other person by any means during or following your exam. You are not allowed to copy/paste text to or from external sources unless this has been authorised by your Chief Examiner.

You must comply with any instructions given to you by Monash exam staff.

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Failure to comply with the above instructions, or attempting to cheat or cheating in an assessment may constitute a breach of instructions under regulation 23 of the Monash University (Academic Board) Regulations or may constitute an act of academic misconduct under Part 7 of the Monash University (Council) Regulations.

Authorised Materials

CALCULATORS	☐ YES	☒ NO
DICTIONARIES	☐ YES	☒ NO
NOTES	☐ YES	☒ NO
WORKING SHEETS	☒ YES	☐ NO
PERMITTED ITEM	☐ YES	☒ NO

if yes, items permitted are:

Instructions

Once your exam finishes, you will be given time to scan a QR code and upload your answers using

 your smartphone and laptop.

[Here's how to do it.](#) 

- There are two types of questions: "short answer questions" and "hand-written response".
- Unless otherwise specified, each answer to a "short answer question" on this exam is a rational number; enter your answers as follows. If the answer is an integer then enter it as a numeral. For example 16, 1, or -1 but NOT sixteen, 1.0, minus 1. If the answer is a non-integer rational number then enter it as [numerator]/[denominator] where the fraction is in lowest terms, [numerator] and [denominator] are integers entered as above. For example, -1/2, 5/4 or 11/16 NOT 1/-2, 1.25 or 22/32. Importantly, no answer to a "short answer question" should contain a space.
- The hand-written responses will require you to upload your written work using a camera (via a phone or tablet) in the 30 minutes after the exam is over. The procedure for doing this will be explained to you at the end of the exam.
- Note: Please widen your browser window to see the questions displayed in full size.
- A formula sheet is provided within the exam, before the questions (on page 2).
- You are not allowed to consult books or online sources during the exam. You may use an unlimited number of blank A4 paper as working sheets.
- Once the exam duration is finished, your exam will automatically submit. Please ensure you finalise your answers before the end of the allocated exam time.

When your exam duration has ended, you are not permitted to write or edit any of your answer sheets. Please note you must remain under exam conditions and are not permitted to adjust your answer sheets during the upload process once you have submitted your exam.

In this assessment, you must not use generative artificial intelligence (AI) to generate any materials or content in relation to the assessment task.

Instructions

Information

You can review your exam instructions by clicking the 'Show Instructions' button above.

Information

Getting help and support

Assessments without online supervision

If you have any difficulties (technical, medical or otherwise), call the Assessment Support hotline.

If your internet connection drops out during your assessment, don't panic – your responses are automatically saved every 30 seconds.



Assessment Support

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Raise hand





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Formula Sheet

Information

$$a^0 = 1, \quad a^x a^y = a^{x+y}, \quad a^{xy} = (a^x)^y.$$

$$\log_a(1) = 0, \quad \log_a(xy) = \log_a(x) + \log_a(y), \quad \log_a(x^y) = y \log_a(x), \quad \log_a(x) = \frac{\log_b(x)}{\log_b(a)}.$$

$$\text{If } f(x) = x^n, \text{ then } f'(x) = nx^{n-1}.$$

$$\text{If } f(x) = a^{cx+d}, \text{ then } f'(x) = c \ln(a) a^{cx+d}.$$

$$\text{If } f(x) = \log_a(cx + d), \text{ then } f'(x) = \frac{c}{\ln(a)(cx+d)}.$$

$$\nabla f(x, y) = \begin{bmatrix} f_x(x, y) \\ f_y(x, y) \end{bmatrix} \quad H(x, y) = \begin{bmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{yx}(x, y) & f_{yy}(x, y) \end{bmatrix}.$$

$$\det(H(x, y)) > 0 \text{ and } f_{xx}(x, y) > 0 \implies \text{local minimum.}$$

$$\det(H(x, y)) > 0 \text{ and } f_{xx}(x, y) < 0 \implies \text{local maximum.}$$

$$\det(H(x, y)) < 0 \implies \text{saddle point.}$$

$$\det(H(x, y)) = 0 \implies \text{test inconclusive.}$$

$$\text{If } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ then } \det(A) = ad - bc \text{ and } A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

	Unordered	Ordered
With repetition	$\binom{n+r-1}{r}$	n^r
Without repetition	$\frac{n!}{r!(n-r)!} = \binom{n}{r}$	$\frac{n!}{(n-r)!}$

Selections of r elements
from a set of n elements:

$$\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

$$\Pr(A|B) = \frac{\Pr(B|A)\Pr(A)}{\Pr(B|A)\Pr(A) + \Pr(B|\bar{A})\Pr(\bar{A})}.$$

$$\text{Discrete uniform: } \Pr(X = k) = \frac{1}{b-a+1} \text{ has } E[X] = \frac{a+b}{2} \text{ and } \text{Var}[X] = \frac{(b-a+1)^2-1}{12}.$$

$$\text{Bernoulli: } \Pr(X = k) = \begin{cases} p & \text{for } k = 1 \\ 1-p & \text{for } k = 0 \end{cases} \text{ has } E[X] = p \text{ and } \text{Var}[X] = p(1-p).$$

$$\text{Geometric: } \Pr(X = k) = p(1-p)^k \text{ has } E[X] = \frac{1-p}{p} \text{ and } \text{Var}[X] = \frac{1-p}{p^2}.$$

$$\text{Binomial: } \Pr(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \text{ has } E[X] = np \text{ and } \text{Var}[X] = np(1-p).$$

$$\text{Poisson: } \Pr(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \text{ has } E[X] = \lambda \text{ and } \text{Var}[X] = \lambda.$$

$$\text{Continuous uniform: } \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise.} \end{cases} \text{ has } E[X] = \frac{a+b}{2} \text{ and } \text{Var}[X] = \frac{(b-a)^2}{12}.$$

$$\text{Exponential: } \lambda e^{-\lambda x} \text{ has } E[X] = \frac{1}{\lambda} \text{ and } \text{Var}[X] = \frac{1}{\lambda^2}.$$

$$\text{Normal: } \frac{1}{\sqrt{2\sigma^2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \text{ has } E[X] = \mu \text{ and } \text{Var}[X] = \sigma^2.$$

$$t\text{-statistic: } t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Foundations

Question 1

Evaluate $\sum_{j=3}^7 j$

25

1.5
Marks

Question 2

Find the value of the following.

$$\prod_{j=2}^4 j^2$$

576

1.5
Marks

Question 3

Choose which of the following terms correctly applies to the function $f(x) = 2|x| + 2$

Select one:

- ☒ a.
Convex
- ☐ b.
Concave
- ☐ c.
Both convex and concave
- ☐ d.
Neither convex nor concave

2
Marks

Question 4

Find the equation of the line through the points $(8, 5)$ and $(6, -7)$. The equation of the line should be of the form $y = mx + b$.

m =

6

b =

-43

2
Marks

Question 5

2

Suppose $f(x) = bx^a$, and that a log-log plot (with logarithms taken base 3) of this function shows a straight line with slope $\frac{2}{3}$ and y-intercept 3. What are a and b ?

a =

-2

b =

27

Functions

Question 6

Consider the function $f : [0, \infty) \rightarrow \mathbb{R}$ where $f(x) = x^2 + 2x$. What is the value of $f^{-1}(24)$?

4

2

Marks

Question 7

If $f(x) = 2(7x + 4)^{3/2}$, find $f'(0)$.

42

2

Marks

Question 8

If $f(x) = (2x^2 - 3)e^{3x}$, find the slope of the tangent line to the graph of f at $x = 0$.

-9

2

Marks

Question 9

Find the maximum value of the product xy , under the constraint that $2x + 3y = 60$.

150

3

Marks

Question 10

For the function $h(x) = e^{2x^2 + 3x - 5}$, find x at which h has a stationary value.

-3/4

2

Marks

Question 11

Evaluate $\int_{-2}^4 (6x^2 + 6x - 4)dx$

156

2

Marks

Question 12

Find the stationary point of the function $f(x, y) = x - \ln(x^2 + y^2)$, with domain $(x, y) \neq (0, 0)$.

Make sure to express the point in the form (a, b) .

(2, 0)

3

Marks

Question 13

For the function $g(x, y) = 2xy(x^2 + y)$, find the gradient vector ∇g at the point $(2, 1)$.

Make sure to express your vector in the form (a, b) .

(26, 24)

3

Marks

Question 14

If the gradient of a function f satisfies

$$\nabla f(x, y) = \begin{pmatrix} 3x^2y^2 + 3x^2y \\ 2x^3y - 2y + x^3 \end{pmatrix},$$

find the determinant of $H(1, -1)$, where H refers to the Hessian matrix.

-9

2

Marks

Question 15

What is the residual sum of squares (RSS) when approximating $(2, 5), (3, 4), (5, 6)$ by $f(x) = 2x + 1$?

3

Marks

34

Linear algebra

Question 16

Let $A = \begin{pmatrix} 2 & 3 \\ 4 & x \end{pmatrix}$. For which $x \in \mathbb{R}$ is A not invertible?

6

2

Marks

Question 17

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & -3 \\ 1 & 0 & -1 \end{pmatrix}$. Then $A^{-1} = \begin{pmatrix} 0 & -1 & x \\ 1/2 & 2 & -9/2 \\ 0 & -1 & 2 \end{pmatrix}$. Find x .

3

2

Marks

Question 18

Let $A = \begin{pmatrix} -2 & -4 & 2 \\ -2 & 1 & 2 \\ 4 & 2 & 5 \end{pmatrix}$. One of the eigenvalues of A is 6; if the corresponding eigenvector is of the form $(1, y, z)$, find y and z .

2

Marks

$y =$ 6

$z =$ 16

Question 19

Let $a = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$ and $b = \begin{pmatrix} x \\ 2 \\ 5 \end{pmatrix}$. Suppose that a and b are orthogonal. What is x ?

-4

2

Marks

Question 20

Find the length of the vector $(4, 3, -12)$.

13

2

Marks

Question 21

3

Marks

Solve the following system of linear equations using Gaussian elimination:

$$\begin{pmatrix} 1 & -1 & 1 \\ 2 & 3 & -1 \\ 3 & -2 & -9 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 \\ -2 \\ 9 \end{pmatrix}$$

x= 4

y= -3

z= 1

Combinatorics

Question 22

How many 3 digit numbers are there which contain only even digits and in which no digits are repeated? (note: such a number cannot start with a 0, however any of the other digits may be zero, since 0 is considered to be even)

48

2

Marks

Question 23

A committee of 7 must be formed out of a group of 10 people. How many different ways are there to form this committee?

120

2

Marks

Question 24

There are 32 letters in the Russian alphabet. Suppose we form a password 9 letters long using only lower case letters from the Russian alphabet. There are exactly 2^n such passwords. Find n .

45

2

Marks

Question 25

At a bakery there are 4 different types of loaves of bread. How many different purchases of 8 loaves can one make?

165

2

Marks

Probability and graphs

Question 26

Random variable X is defined to be the value when a fair six-sided die is rolled, and Y is to be the value when a fair ten-sided die is rolled. Find the expectation $E[2X + 4Y]$.

29

2
Marks

Question 27

A game is played in which a fair coin is tossed three times. If 2 or more heads are thrown the player wins 24 dollars, if one head occurs the player wins 8 dollars and if no heads occur the player loses 16 dollars. What is the expected value of their winnings, in dollars? Note: just put the number, do not put a dollar sign.

13

3
Marks

Question 28

Suppose a continuous random variable X has a probability density function $h(x)$ defined by $h(x) = kx^2(1 - x)$ for $0 < x < 1$, and $h(x) = 0$ for x outside this interval. Find k .

12

3
Marks

Question 29

A graph has a degree sequence of $(1, 2, 2, 2, 3, 3, 3)$. How many edges does it have?

8

2
Marks

Question 30

The graph G has the adjacency matrix $M =$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

How many vertices does G have?

9

How many edges does G have?

6

How many components (i.e. separate pieces) does G have?

3

4
Marks

Question 31

2

Marks

Suppose a random variable X admits the pdf $h(x) = \frac{3}{8}x^2$ for $0 < x < 2$ and 0 elsewhere. Find $E[X]$ and $\text{Var}(X)$; remember to reduce all fractions, and leave numbers as improper fractions rather than mixed numbers or decimals.

$$E[X] = \frac{3}{2}$$

$$\text{Var}(X) = \frac{3}{20}$$

Long answer questions

Question 32

Let $f(x) = 2x^3 - 3x^2 - 12x + 4$ on domain $[-3, 3]$. Find all local extrema of f on its domain, and find also the global extrema of f on its domain. Classify the extrema as either maximums or minimums. Explain your method clearly.

6
Marks



Please answer the question on your blank piece of paper.

- After your exam finishes, you'll have extra time to access your phone to scan a QR code and upload your answer.
- Clearly label each page with Student ID and this question number (and sub part if applicable) (for example, 'Question 7a')
- Do not write your Name on it

No. of answer sheets: 3

Question 33

Let $f(x, y) = 3xy^2 + x^3 - 3x^2 - 3y^2 + 2$. Determine the gradient, i.e. ∇f , and hence find the Hessian matrix of $f(x, y)$. Find the stationary points, and use the Hessian to classify as much as possible the nature of the stationary points of f (note: if the Hessian test fails just say so, you do not need to try other tests).

6
Marks



Please answer the question on your blank piece of paper.

- After your exam finishes, you'll have extra time to access your phone to scan a QR code and upload your answer.
- Clearly label each page with Student ID and this question number (and sub part if applicable) (for example, 'Question 7a')
- Do not write your Name on it

No. of answer sheets: 3

Question 34

Consider the following matrix.

$$A = \begin{bmatrix} -3 & 5 \\ 6 & -2 \end{bmatrix}$$

6
Marks

- Find the eigenvalues of A .
- Find eigenvectors corresponding to each of the eigenvalues.
- Find matrices P, D with P invertible and D diagonal so that $A = PDP^{-1}$. You do not need to calculate P^{-1} .



Please answer the question on your blank piece of paper.

- After your exam finishes, you'll have extra time to access your phone to scan a QR code and upload your answer.
- Clearly label each page with Student ID and this question number (and sub part if applicable) (for example, 'Question 7a')
- Do not write your Name on it

No. of answer sheets: 3

Question 35

Consider the following system of linear equations in matrix form:

$$\begin{bmatrix} 1 & -2 & 2 & -1 \\ 3 & 1 & 6 & 11 \\ 2 & -1 & 5 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 3 \\ 16 \\ 10 \end{bmatrix}$$

Solve the system and write down all solutions (x, y, z, t)

6

Marks



Please answer the question on your blank piece of paper.

- After your exam finishes, you'll have extra time to access your phone to scan a QR code and upload your answer.
- Clearly label each page with Student ID and this question number (and sub part if applicable) (for example, 'Question 7a')
- Do not write your Name on it

No. of answer sheets: 3

Question 36

A commuter walks 10% of the time and uses her car the remaining 90% of the time. She is late 60% of the time when she walks, but she is late only 20% of the time when she drives.

- a) What is the probability that she walked if she was late?
- b) What is the probability that she drove if she is on time?

6

Marks



Please answer the question on your blank piece of paper.

- After your exam finishes, you'll have extra time to access your phone to scan a QR code and upload your answer.
- Clearly label each page with Student ID and this question number (and sub part if applicable) (for example, 'Question 7a')
- Do not write your Name on it

No. of answer sheets: 3

32. $f'(x) = 6x^2 - 6x - 12 = 6(x^2 - x - 2)$
 $= 6(x-2)(x+1)$, so we have stationary
pts. at $x=2, -1$. $f''(x) = 12x - 6$, and
 $f''(2) > 0$, $f''(-1) < 0$, so $x=2$ is a local min,
 $x=-1$ is a local max. To find the global
extrema, we must try these points, as well as
the endpoints of the interval. We have
 $f(2) = -16$, $f(-1) = 11$, $f(-3) = -41$, $f(3) = -5$,
so global max is at $(-1, 11)$, global min is $(-3, -41)$.

$$33. \nabla f = \begin{pmatrix} 3y^2 + 3x^2 - 6x \\ 6xy - 6y \end{pmatrix} = \begin{pmatrix} 3(x^2 - 2x + y^2) \\ 6(x-1)y \end{pmatrix}$$

Stationary pts. require $\nabla f = 0$, so we must have
 $6(x-1)y = 0$, thus $x=1$ or $y=0$. If $y=0$, then
 $x^2 - 2x = 0$, so $x=0$ or $x=2$. If $x=1$, then
 $y^2 - 1 = 0$, so $y = \pm 1$. We have then 4 stationary
pts, $(0, 0)$, $(2, 0)$, $(1, 1)$, $(1, -1)$. The Hessian
matrix is

$$H(x, y) = \begin{pmatrix} 6x - 6 & 6y \\ 6y & 6x - 6 \end{pmatrix}$$

So $\det(H(x,y)) = 36((x-1)^2 - y^2)$. This is positive at $(0,0)$ and $(2,0)$, and negative at $(1,1)$, $(1,-1)$. Thus, $(1,1)$ and $(1,-1)$ are saddle pts., and $(0,0)$, $(2,0)$ are local extrema. Since $f_{xx}(0,0) < 0$, this point is a local max, and $f_{xx}(2,0) > 0$ so this is a local min.

$$34. a) \det(A - \lambda I) = (-3 - \lambda)(-2 - \lambda) - 30 \\ = \lambda^2 + 5\lambda - 24 = (\lambda + 8)(\lambda - 3)$$

So $\lambda = -8, 3$ are e. values.

$$b) (A - 3I) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -6 & 5 \\ 6 & -5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -6x + 5y \\ 6x - 5y \end{pmatrix}.$$

If this is $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, then $5y = 6x$, and $v_1 = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$ works.

$(A + 8I) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 & 5 \\ 6 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5x + 5y \\ 6x + 6y \end{pmatrix}$. If this is $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, then $x = -y$, and $v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ works. Each of these may be multiplied by any scalar and it is still correct.

34 c) As per Lecture 12, if we let $P = (v_1/v_2)$
 then $PAP^{-1} = \begin{pmatrix} 3 & 0 \\ 0 & -8 \end{pmatrix}$, where $P = \begin{pmatrix} 5 & 1 \\ 6 & -1 \end{pmatrix}$.

35.
$$\left(\begin{array}{cccc|c} 1 & -2 & 2 & -1 & 3 \\ 3 & 1 & 6 & 11 & 16 \\ 2 & -1 & 5 & 4 & 10 \end{array} \right) \rightarrow \left(\begin{array}{cccc|c} 1 & -2 & 2 & -1 & 3 \\ 0 & 7 & 0 & 14 & 7 \\ 0 & 3 & 1 & 6 & 4 \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 1 & -2 & 2 & -1 & 3 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 3 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right)$$

we take t as free variable. Then $z=1$,
 $y+2t=1$, and $x+3t=3$. The soln is $(3-3t, 1-2t, 1, t)$.

36. a) We have $P(L|W) = .6$, so by Bayes' Thm.

$$P(W|L) = \frac{P(L|W)P(W)}{P(L)} = \frac{.6 \times .1}{.6 \times .1 + .2 \times .9} = \frac{.6}{.24} = \frac{1}{4}.$$

b) We have $P(T|D) = .8$, so

$$P(D|T) = \frac{P(T|D)P(D)}{P(T)} = \frac{.8 \times .9}{.4 \times .1 + .8 \times .9} = \frac{.72}{.76} = \frac{18}{19}.$$

They do not need to reduce fractions here,
 any form is OK.